

Problem 1-Direct proportion:

Force (F) 45 Newtons (GIVEN)

Extension (x) = ? cm

$$\frac{F}{x} = k$$

$$k = 15 \quad (GIVEN)$$

$$x \times 15 = 45$$

$$\Rightarrow \frac{x \times 15}{15} = \frac{45}{15}$$

$$\Rightarrow x = 3 \text{ cm}$$

(Only if you know the units for 'k')

If $k = 15 \text{ N/cm}$, then $x = 3 \text{ cm}$

If $k = 15 \text{ N/m}$, then $x = 3 \text{ m}$
 $= 3 \times 100 \text{ cm}$
 $= 300 \text{ cm}$

Problem 2: Direct proportion

For $\frac{CF}{x} = 20$ Newtons (GIVEN)

Extension $x = 4$ cm (GIVEN)

$$k = \frac{F}{x}$$

$$= \frac{20}{4}$$

$$= 5 \text{ N/cm}$$

Problem 3—Direct proportion:

$$\underline{\underline{F}} \propto x \quad \text{stretch} \quad 10 \text{ cm} = 20 \times \frac{10}{4} \text{ cm}$$

$$= \frac{200}{4} \text{ cm}$$

$$= 10 \times \frac{20}{4} \text{ cm}$$

$$= 10 \times 5 \text{ cm}$$

$$= 50 \text{ cm}$$

50
N

$$\frac{F}{10 \text{ cm}} = \frac{20 \text{ N}}{4 \text{ cm}}$$

$$\Rightarrow F = \frac{5}{20 \text{ N}} \times 10 \text{ cm}$$
$$= 50 \text{ N}$$

Problem 4—Direct proportion:

$$V = R \times I \quad (\text{or}) \quad \frac{V}{I} = R$$

$$\Rightarrow \frac{V}{I} \propto R \quad V \propto I \quad (\propto) \quad \frac{V}{I} = R$$

$$R = 12 \text{ Ohms}$$

$$I = 3 \text{ Amps}$$

$$V = R \times I$$

$$= 12 \times 3$$

$$= 36 \text{ Volts}$$

$$V = IR$$

$$\text{volts} = \text{amperes} \times \text{ohms}$$

$$\text{ohms} = \frac{\text{volts}}{\text{amperes}}$$

Problem 5-Direct proportion:

Current increases by the
same rate.

If voltage triples, current also triples
(only if 'R' is constant)

$$\checkmark V = \checkmark I \times R =$$

Problem 6 - Direct proportion:

$$I \times R = V$$

$$\Rightarrow 6 \times 4 = 24$$

$$x \times 2 = 24$$

$$\Rightarrow \frac{x \times 2}{2} = \frac{24}{2}$$

$$\Rightarrow x = 12$$

voltage is
not
remaining
the same
Resistance
is!

$$\begin{aligned} \text{New resistance} &= R \\ &= \text{old resistance} \\ &= 4 \text{ Ohms} \end{aligned}$$

$$\text{New current } I = 2 \text{ A}$$

New voltage

$$V = I \times R$$

$$= 2 \text{ A} \times 4 \text{ Ohms}$$

$$= 8 \text{ Volts}$$

Problem 7 - Direct proportion:

2 cm double stretch of the spring for every 5 N force.

$$F = k \times x$$

$$5 = k \times 2$$

$$\frac{5}{2} = k \times \frac{2}{2}$$

$$2.5 = k$$

Table:

5 Newtons	10 Newtons	15 Newtons	20 N
2 cm	4 cm	6 cm	8 cm

$$x = \frac{1}{2.5} \times F$$

$$= \frac{1}{2.5} \times F$$

$$= \frac{10}{25} \times F$$

$$x = \frac{2}{5} \times F$$

Problem 8 - Direct proportions:

I, I, I increase V by the
same rate amount

— double the Voltage

$$V = I \times R \quad ; \quad V \text{ \& } I \text{ are the initial values}$$

If 'R' is constant & you increase I
by 10 A, how much is new V?

$$I_{\text{new}} = I + 10$$

Here 'V' didn't
increase by
10 V

$$V_{\text{new}} = I_{\text{new}} \times R = (I + 10) \times R$$

If 'R' is constant & you double I,
how much is new V?

$$I_{\text{new}} = 2 \times I$$

$$V_{\text{new}} = I_{\text{new}} \times R = 2 \times I \times R = 2 \times V \rightarrow \text{so 'V' also doubled}$$

Problem 9: $\propto$?

$$x \propto y$$

$$\Rightarrow 1 \propto 8 = y$$

$$\Rightarrow 8 \propto y$$

$$\Rightarrow y = 8 \propto$$

$$y \propto x$$

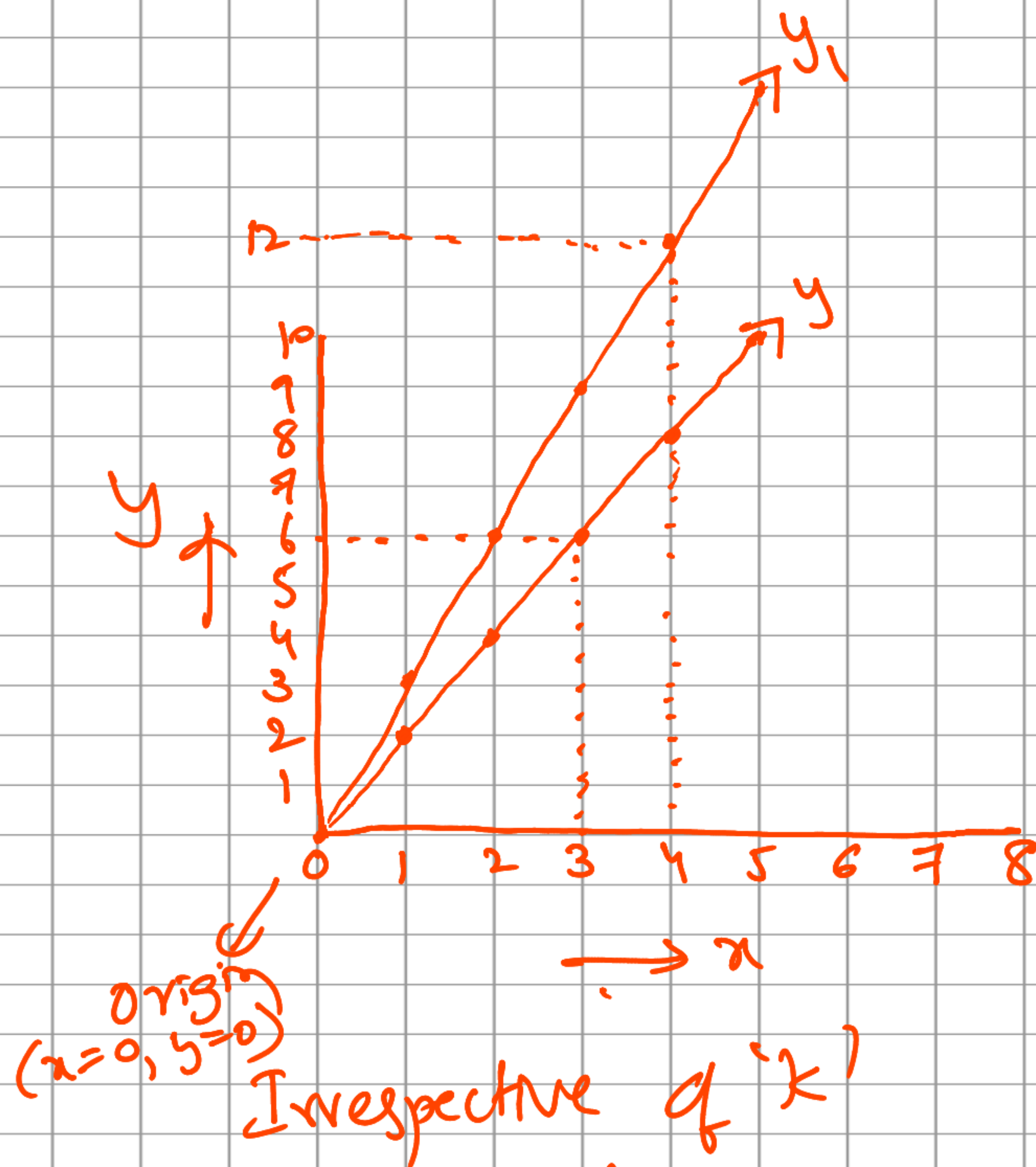
$$\Rightarrow \frac{y}{x} = k$$

$$\Rightarrow k = \frac{24}{3} = 8 \quad \checkmark$$

$$\text{If } x = 10, \quad y = k \times x = 8 \times 10 = 80$$

✓

problem 10: I don't know about graphs 😊!



Irrespective of 'k'

$$y = kx$$

means $y=0$ if $x=0$,

so the line passes through ($x=0, y=0$) = origin always.

$$y \propto x$$

$$y = kx$$

Let $k=2$ $x=3$

x	$y = 2x$	$y = 3x$
0	0	0
1	2	3
2	4	6
3	6	9
4	8	12
5	10	15

problem 11- Indirect proportion:

$$m \times a = k$$

$$\Rightarrow 5 \times 20 = 100$$

$$\Rightarrow 100 = k$$

$$\Rightarrow k = 100$$

Where are the descriptions?
What's 'm'? What's 'a'?

$$\text{mass 'm'} = 5 \text{ kg}$$

$$\text{acceleration 'a'} = 20 \text{ m/s}^2$$

$$m \propto \frac{1}{a}$$

$$\Rightarrow m \times a = F$$

$$\Rightarrow F = m \times a = 5 \text{ kg} \times 20 \text{ m/s}^2 = 100 \text{ kg} \cdot \frac{\text{m}}{\text{s}^2} = 100 \text{ N}$$

Problem 12-Indirect proportion:

1 f m d o u b (e s , a c u t s i n h a l f .

$$k = (m \times 2) \times \left(\frac{a}{2}\right)$$

$$\Rightarrow k = (5 \times 2) \times \left(\frac{20}{2}\right)$$

$$\Rightarrow k = 10 \times 10$$

$$\Rightarrow k = 100 ; \quad a = 10$$

$m \times a$ is constant (or) $m_1 a_1 = m_2 a_2$

$$\Rightarrow 5 \times 20 = 10 \times a_2$$

$$\Rightarrow a_2 = \frac{5 \times 20}{10} = 10 \frac{m}{s^2}$$

problem 13 - indirect proportion:

$$k = P \times v$$

$$\Rightarrow k = 1 \times 0 \times 2$$

$$\Rightarrow k = 2 \times 0 \quad \text{litres} \times \text{atom.}$$

problem 14 - Indirect proportion:

I need help.

$$V \propto \frac{1}{P} \text{ (or)}$$

$$V \times P = \text{constant}$$

$$\text{(or)} \quad V_1 \times P_1 = V_2 \times P_2$$

$$10 \times 2 = 4 \times P_2$$

$$\Rightarrow P_2 = \frac{10 \times 2}{4} = 5 \text{ atm.}$$

problem 15- Indirect proportion:

$$x \times y = k$$

$$k = 10 \times 2$$

$$= 20$$

If $x = 5$, what is y ?

$$y = \frac{k}{x}$$

$$\Rightarrow y = \frac{20}{5} \quad \text{or} \quad \frac{20}{5}$$

$$\Rightarrow y = 4$$

problem 16 - Indirect proportion:

I t d o u b l e s . I n e e d h e l p .

$$m_1 a_1 = m_2 a_2$$

$$m_2 = \frac{m_1}{2}$$

$$\cancel{m_1} a_1 = \frac{\cancel{m_1}}{2} \times a_2$$

$$\Rightarrow a_2 = 2 \times a_1$$

problem 17 Indirect proportion

1 neeo help.

$$P_{\text{bottom}} V_{\text{bottom}} = P_{\text{top}} V_{\text{top}}$$

$$\text{If } P_{\text{top}} < P_{\text{bottom}} \Rightarrow k = \frac{P_{\text{bottom}}}{P_{\text{top}}} > 1.$$

$$\Rightarrow V_{\text{top}} = \frac{P_{\text{bottom}}}{P_{\text{top}}} \times V_{\text{bottom}}$$

$$\Rightarrow V_{\text{top}} = k \times V_{\text{bottom}}$$

Where $k > 1$

$$\Rightarrow V_{\text{top}} > V_{\text{bottom}}$$

Problem 18:

You have cut it into a
quarter.

$$a_2 = 4 \times a_1$$

$$m_1 a_1 = m_2 a_2$$

$$m_1 \cancel{a_1} = m_2 \times 4 \times \cancel{a_1}$$

$$\Rightarrow m_2 = \frac{m_1}{4}$$

Problem 19:

P g e t s v e r y s m a l l .

close to zero

Problem 20:

k i n $\propto$ y i s $\frac{x}{y}$

k i n $x \propto \frac{1}{y}$ i s $x x y$.